Prithvi Foundation Model

NASA × IBM

Architecture · Training Data · Applications · Compute Requirements

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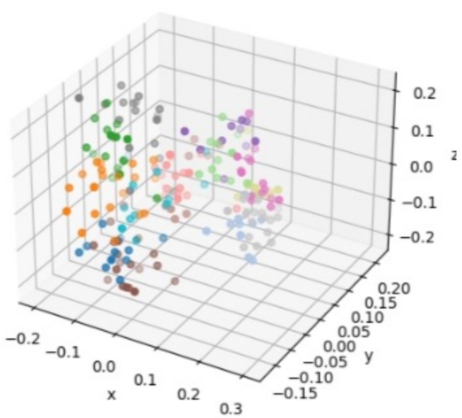
01

Key Findings



Architecture	Prithvi is a Vision Transformer adapted to read multi-date satellite images by learning patterns across time and space.
Training Data	Trained on 4.2 million satellite image sequences from NASA's Landsat and Sentinel-2 satellites, covering over 800 ecosystem types around the world across 7 years.
Applications	Wildfire scar mapping, flood detection, crop monitoring, and landslide detection.
Compute	The smaller 300M model runs on a free Google Colab GPU. Fine-tuning needs a more powerful GPU, but lightweight techniques like LoRA cut that cost significantly.

02

Open Questions

1	**Resolution limit**	Prithvi's training images are 30 m per pixel. Will that be detailed enough for fine-grained tasks, or do we need higher-resolution data?
2	**Regional gaps**	The model was trained globally, but some regions have much less coverage. How well does it work in areas it has seen less of, like Morocco's Middle Atlas?
3	**Label efficiency**	How much labeled data is needed for fine-tuning?
4	**Time sensitivity**	Can it capture fast-changing events like fires?
5	**Location awareness**	Prithvi includes location-based embeddings. How much does knowing where a tile is (latitude/longitude) actually improve predictions?

03 Recommendations

01
Start small
Run the 300M model on Google Colab (free T4 GPU) first. Make sure the Sentinel-2 bands load correctly before doing anything more complex.

02
Gather data
Download the Sentinel-2 tiles, ERA5 climate data, SRTM elevation, and SYSFEU fire records. Expect to need 50–500 GB of storage.

03
Use LoRA
When fine-tuning, use LoRA — a technique that adapts the model using far less memory (~8 GB vs. 80 GB). It makes the 300M model trainable on a single consumer GPU.

04
Pin versions
Lock your software versions (TerraTorch, PyTorch Lightning) on day one. These tools are updated frequently and small changes can break your pipeline.

05
Benchmark
Compare your results against the existing RICER XGBoost model. Use standard metrics (F1, AUC) so the comparison is fair and reproducible.

04

Why Prithvi for RICER?

Wildfire Domain Match

Prithvi was already fine-tuned to map wildfire burn scars from space. Our RICER task is closely related. That means Prithvi arrives with relevant knowledge built in, rather than starting from zero.

No Data Reformatting Needed

RICER uses Sentinel-2 satellite images. Prithvi was trained on HLS data — a product that blends Landsat and Sentinel-2 into a common format using the same 6 spectral bands. We can feed our images directly into the model.

Label Efficiency

Historical fire records for Morocco's Middle Atlas are sparse. Foundation models like Prithvi were designed to work well with limited labeled data — making them a natural fit when ground-truth labels are hard to come by.

05

RICER Pipeline

01

Collect

Download Sentinel-2 images, ERA5 climate data, SRTM elevation, and SYSFEU fire records for Morocco's Middle Atlas.

02

Embed

Send each Sentinel-2 image tile through Prithvi. The model outputs a compact list of numbers (an embedding) that summarizes what it sees.

03

Fuse

Combine Prithvi embeddings with climate and terrain data into one table — one row per location, ready for machine learning.

04

Predict

Train XGBoost on the fused table. It outputs a risk score for each location: Very Low / Low / Moderate / High / Very High.

05

Compare

Test three versions: original XGBoost, improved XGBoost, and Prithvi + XGBoost. See which predicts fire risk most accurately.

References

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